

DevOps Training

Course Curriculum

Learn • Build • Automate • Deploy

Format Instructor-Led Training + Practical Demos	Duration 1 Month
Prerequisite Linux fundamentals	Focus DevOps tools, CI/CD & automation

Program Overview

This DevOps Program provides in-depth knowledge of DevOps tools including Git, Jenkins, Docker and Ansible, with a focus on CI/CD best practices and continuous monitoring across the software development lifecycle.

Training includes live online classroom sessions, real-world projects, whiteboard explanations, practical demonstrations and flexible weekday/weekend delivery.

Structured Learning Path

The curriculum below retains the modules and terminology from the provided DevOps training curriculum while presenting them in the NextGen DevSecOps AI format.

MODULE 01

Waterfall Model (Traditional)

- Traditional software delivery approach and its characteristics.

MODULE 02

DevOps Essentials

- Why DevOps?
- What is DevOps?
- DevOps market trends
- What is a pipeline?
- CI, Continuous Delivery and Continuous Deployment concepts
- General CI/CD workflow
- Common challenges when adopting DevOps principles

MODULE 03

Introduction to Agile

- Agile methodology
- Daily stand-ups

MODULE 04

GitHub Branching Strategy

- Branching strategy and collaborative source-control practices

MODULE 05

Developer Workflow on GitHub

- Developer workflow and collaboration using GitHub

MODULE 06

Managing Source Control with Git

- What is version control
- Centralized and distributed version control
- What is Git?
- Developer/coding workflows
- Branching and merging concepts
- Working with remote repositories
- Practical demonstration

MODULE 07

Maven Build Tool

- Overview of build tools
- What is Maven?
- Maven plugins
- Maven archetypes
- Project Object Model (POM)
- Source-control integration
- Maven build lifecycle
- Practical demonstration

MODULE 08

Virtual Machines & Jenkins Setup

- Setup virtual machines using VMware
- Jenkins Master and Slave node setup

MODULE 09

Continuous Integration with Jenkins

- What is Continuous Integration?
- Why CI is required
- Introduction to Jenkins
- Jenkins architecture
- Adding a slave node to Jenkins Master
- Declarative pipeline using Pipeline as Code (Jenkinsfile)
- Webhooks
- Practical demonstration

MODULE 10

Containerization using Docker

- Introduction to Docker and containers
- Understanding images and containers
- Docker fundamentals
- Docker networking
- Container lifecycle
- Practical demonstration

MODULE 11

Continuous Deployment on Docker using Jenkins

- Continuous deployment workflow using Jenkins and Docker
- Practical demonstration

MODULE 12

Kubernetes — Optional

- Basics of Kubernetes container orchestration
- Docker Swarm vs Kubernetes

- Kubernetes architecture
- Installing Kubernetes using Kubeadm
- Creating Pods and Deployments using YAML
- Selectors and Labels
- ReplicaSets and rolling updates
- Application scheduling
- Services in Kubernetes
- Practical demonstration

MODULE 13

Configuration Management with Ansible

- Overview of configuration management
- Introduction to Ansible
- Ansible architecture
- Ansible components
- Installation and configuration
- Writing Ansible playbooks
- Working with Ansible variables
- Working with Ansible modules
- Practical demonstration

CAPSTONE PROJECT

Build • Integrate • Deploy

The source curriculum defines a practical capstone focused on building and deploying a Java application through Maven, Jenkins and Docker.

Project Component	Hands-on Outcome
Maven	Build and compile a Java project using Maven.
Jenkins Freestyle Job	Create a Jenkins job to automate the Maven build.
Jenkins Declarative Pipeline	Build the Java application using a Docker image.
Docker Hub	Build the image and push it to Docker Hub.

Training Features

- Live online classroom training by an experienced instructor
- Real-world projects and practical hands-on experience
- Whiteboard explanations and practical demonstrations
- Flexible weekday or weekend class timings
- Tailored modules focused on practical DevOps skills

Prerequisites & Audience

Prerequisite: learners should have a technical background with an understanding of Linux fundamentals.

Target audience: software developers, technical project managers, architects, operations support, deployment engineers, freshers and induction batches.

NEXTGEN DEVSECOPS AI

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